AI Usage DISCLOSURE FORM

1. **Team Details**

Team Name: Team Atomic Claws

Project / Product Name: Project Sentinel

Organization / Institution: R.V. College of Engineering, Bangalore (RVCE)

Submission Date: 05-05-2026

2. **AI Usage Declaration**

Did your team use any Artificial Intelligence (AI) in developing this project? Yes / No

YES

3. **Purpose of AI Usage (Brief Details)**

Idea generation / brainstorming - architecture choices redo-log IPC vs pipes or sockets, rolling window size for forecasting, Z-score vs IQR for anomaly detection

Code generation or assistance - Generated C++ engine, Python bridge, anomaly detector, REST API, web dashboard, OpenClaw skill, test suite (157 tests)

UI / UX design - Generated web dashboard (sentinel_web.html) with dark theme, gauge bars, sparklines, mode timeline, SSE streaming

Content creation - Generated README, SUBMISSION.md, SOUL.md, SKILL.md

Data analysis - Generated statistical analysis modules linear regression forecaster, Z-score anomaly detector, rolling variance ratio

Testing / debugging - Generated 157 tests across 5 test files; debugged Windows file-locking issue, CRLF handling, git merge conflict

Other - Generated OpenClaw workspace files (SOUL.md, AGENTS.md, HEARTBEAT.md, SKILL.md) for hackathon compliance

4. **Feature Origin**

1. Feature Name - Multi-Signal Hardware Fusion Engine

2. Self-Generated/AI-Generated/Both - Both (Self-Generated concept + AI-Generated code)

3. Description -

- **AI Tool:** Claude Sonnet 4.6 via Claude.ai
- **Prompt:** *"Design a hardware telemetry bridge between a C++ polling engine and a Python AI agent using an append-only redo log as IPC. The bridge should combine RAM, CPU, thermal, battery, and disk into a composite pressure state."*
- **Output Summary:** C++ engine (sentinel_engine.cpp) + Python engine (sentinel_engine_py.py) + bridge parser (sentinel_bridge.py) with ContextAnalyser class
- **Modification:** Team defined the threshold values (RAM 75%/90%, temp 70°C/85°C), the LSN stamping design, and the decision to use linear regression (not ML) for forecasting

1. Feature Name - Statistical Anomaly Detector (8 Patterns)

2. Self-Generated/AI-Generated/Both – Both (Self-Generated concept + AI-Generated code)

3. Description –

- **AI Tool:** Claude Sonnet 4.6 via Claude.ai
- **Prompt:** *"Implement statistical anomaly detection for hardware metrics using Z-score for spike detection, linear regression for creep detection, and variance ratio for oscillation detection. Return typed Anomaly objects with severity and narrative."*
- **Output Summary:** anomaly_detector.py - 8 failure-signature patterns: RAM_SPIKE, RAM_CREEP, THERMAL_RUNAWAY, BATTERY_CLIFF, CPU_THRASH, COMPOUND_PRESSURE, MEMORY_OSCILLATION, RECOVERY_DETECTED
- **Modification:** Team identified which patterns matter for Galaxy devices (thermal runaway, battery cliff), defined severity thresholds, validated against real hardware behaviour

1. Feature Name - Adaptive AI Agent with Per-Turn Hardware Injection

2. Self-Generated/AI-Generated/Both – Both (Self-Generated concept + AI-Generated code)

3. Description –

- **AI Tool:** Claude Sonnet 4.6 via Claude.ai
- **Prompt**: *"Write an AI agent that re-samples hardware state and re-injects it into the Claude system prompt before every conversation turn. Map pressure states to operating modes: FULL (1024 tokens), QUANTIZED (512), MINIMAL (256)."*
- **Output Summary**: sentinel_agent.py - FULL/QUANTIZED/MINIMAL/SUSPEND modes, dynamic token budget, echo mode for testing

- **Modification:** Team defined the mode-switching logic, designed the proactive suspension behaviour, and validated token budget values against Claude API response quality

1. Feature Name - OpenClaw Skill Integration

2. Self-Generated/AI-Generated/Both – Both (Self-Generated concept + AI-Generated code)

3. Description –

- **AI Tool:** Claude Sonnet 4.6 via Claude.ai
- **Prompt:** *"Write an OpenClaw SOUL.md that gives an AI agent a hardware-aware personality with strict adaptive mode rules - FULL, QUANTIZED, MINIMAL, SUSPEND - based on device resource pressure. Also write a custom Skill with before_turn, session_start, and after_turn triggers."*
- **Output Summary:** SOUL.md, AGENTS.md, HEARTBEAT.md, IDENTITY.md, SKILL.md, run.py - complete OpenClaw workspace
- **Modification:** Team designed the trigger architecture, defined what hardware events get logged to memory, and tested all three trigger handlers

1. Feature Name - Session Hardware Memory

2. Self-Generated/AI-Generated/Both – Both (Self-Generated concept + AI-Generated code)

3. Description –

- **AI Tool:** Claude Sonnet 4.6 via Claude.ai
- **Prompt:** *"Build a session memory system that tracks hardware events across a conversation — mode changes, threshold crossings, anomaly events, peak stats. It should answer hardware history questions directly from memory without an API call."*
- **Output Summary:** sentinel_memory.py — SessionMemory, HardwareEvent, MemoryMixin classes; direct-answer system for hardware questions
- **Modification:** Team identified the dedup bug (mismatched dictionary keys between _add() and _should_record()), fixed it, and designed the event deduplication window

1. Feature Name - Live Web Dashboard + REST API

2. Self-Generated/AI-Generated/Both – AI-Generated code

3. Description –

- **AI Tool:** Claude Sonnet 4.6 via Claude.ai
- **Prompt:** *"Create a self-contained HTML dashboard with live SSE streaming, animated gauge bars, sparklines for 30-second history, mode timeline, anomaly badges, and forecast text. Auto-connect to API; fall back to built-in demo if server unavailable."*

- **Output Summary:** sentinel_web.html (dashboard) + sentinel_api.py (REST API with /status, /stream, /anomalies, /predict, /summary endpoints)
- **Modification:** Team added the scenario simulation buttons, tuned the colour thresholds, and fixed the git merge conflict that broke the API file

5. **Ethical & Compliance Confirmation**

AI usage complies with guidelines and policies. **Yes**

No proprietary or copyrighted data misused. **I Agree**

6. **Declaration & Sign-Off**

Name of Team Representative: Kavya Jain

Role: Developer and Team Lead

Signature: Kavya Jain

Date: 05-05-2026